

EXERCISE 2.10

Q.1 Find two positive integers whose sum is 30 and their product will be maximum.

Solution:

Let x and $30 - x$ be two positive integers. Their product is $x(30 - x)$

$$\begin{aligned}\text{Let } f(x) &= x(30 - x) \\ &= 30x - x^2\end{aligned}$$

Differentiate w.r.t " x ".

$$f'(x) = 30 - 2x \text{ and } f''(x) = -2$$

$$\text{put } f'(x) = 0$$

$$30 - 2x = 0 \Rightarrow x = 15$$

$$\text{At } x = 15 \quad f''(x) = -2 < 0$$

so f will be maximum

So the required two positive integers are 15 and 15.

Q.2 Divide 20 into two parts so that the sum of their squares will be minimum.

Solution:

Let the required two parts are x and $20 - x$ then the sum of their squares will be $x^2 + (20 - x)^2$

$$\text{Let } f(x) = x^2 + (20 - x)^2$$

Differentiate w.r.t " x ".

$$f'(x) = 2x + 2(20 - x)(-1)$$

$$\text{Put } f'(x) = 0$$

$$2x - 2(20 - x) = 0$$

$$2x = 2(20 - x)$$

$$x = 10$$

$$\text{at } x = 10 \quad f''(x) = 4 > 0$$

so $f(x)$ will be minimum

so one of the integers is 10

and other one is $20 - 10 = 10$.

So the required two integers are 10 & 10.

Q.3 Find two positive integers whose sum is 12 and the product of one with the square of the other will be maximum.

Solution:

Let two integers are x and $12 - x$ then the product of $12 - x$ and the square of x is $x^2(12 - x)$

$$\text{Let } f(x) = x^2(12 - x)$$

$$f(x) = 12x^2 - x^3$$

Differentiate w.r.t. ' x '

$$f'(x) = 24x - 3x^2 \text{ and } f''(x) = 24 - 6x$$

$$\text{Put } f'(x) = 0$$

$$x(24 - 3x) = 0$$

$$x = 0, x = 8$$

$$\text{at } x = 0, f''(0) = 24 > 0$$

So f will be minimum so we discard this possibility

$$\text{At } x = 8, f''(8) = 24 - 48 = -24 < 0$$

So f will be maximum

Hence the required integers are

$$x = 8$$

$$12 - x = 12 - 8 = 4$$

Q.4 The perimeter of a triangle is 16cm. If one side is of length 6cm, what are lengths of the other sides for maximum area of the triangle?

Solution:

Sum of lengths of unknown sides is $16 - 6 = 10\text{cm}$

Let the lengths of unknown sides be x and $10 - x$.

If A is the area of triangle, then

$$s = \frac{6 + x + 10 - x}{2} \Rightarrow s = 8$$

$$A = \sqrt{(8)(8 - 6)(8 - x)(8 - 10 + x)}$$

By Hero's formula

$$A = \sqrt{(8)(2)(8 - x)(-2 + x)}$$

$$A = 4\sqrt{10x - x^2 - 16}$$

The maximum value of A depends on the function

$$f(x) = 10x - x^2 - 16$$

Differentiate w.r.t “ x ”

$$f'(x) = 4 \frac{1}{2\sqrt{10x - x^2 - 16}} (10 - 2x)$$

$$\frac{dA}{dx} = \frac{20 - 4x}{\sqrt{10x - x^2 - 16}}$$

Put $\frac{dA}{dx} = 0$ gives,

$$\frac{20 - 4x}{\sqrt{10x - x^2 - 16}} = 0 \Rightarrow x = 5$$

Now

$$\frac{d^2A}{dx^2} = 4 \frac{\sqrt{10x - x^2 - 16}(-1) - (5 - x) \frac{10 - 2x}{2\sqrt{10x - x^2 - 16}}}{10x - x^2 - 16}$$

at $x = 5$,

$$\left. \frac{d^2A}{dx^2} \right|_{x=5} = \frac{-36}{(50 - 25 - 16)^{3/2}} = -\frac{4}{3} < 0$$

Since $\left. \frac{d^2A}{dx^2} \right|_{x=5} < 0$

So A is maximum

So the lengths of unknown sides are 5 and $10 - 5 = 5$

Q.5 Find the dimensions of a rectangle of largest area having perimeter 120cm.

Solution:

Let x be the length of rectangle and y be breadth of rectangle
then perimeter $2x + 2y = 120$

$$\text{or } x + y = 60$$

$$\text{or } y = 60 - x$$

Let A be the area of rectangle,
then $A = xy$

$$\text{or } A = x(60 - x)$$

$$A = 60x - x^2$$

Differentiate w.r.t. x

$$\frac{dA}{dx} = 60 - 2x$$

$$\text{Put } \frac{dA}{dx} = 0 \Rightarrow 60 - 2x = 0$$

$$x = 30$$

$$\text{And } \frac{d^2A}{dx^2} = -2$$

$$\text{At } x = 30 \frac{d^2A}{dx^2} = -2 < 0,$$

So A is maximum

So length of rectangle is $x = 30$ and
breadth $60 - x = 30$

Q.6 Find the lengths of the sides of a variable rectangle having area 36cm^2 when its perimeter is minimum.

Solution:

Let x and y be the lengths of sides of rectangle, then

$$\text{Area} = xy = 36 \Rightarrow y = \frac{36}{x}$$

If p denotes the perimeter of rectangle then

$$p = 2x + 2y$$

$$p = 2\left(x + \frac{36}{x}\right) \quad \left(\because y = \frac{36}{x}\right)$$

Differentiate w.r.t “ x ”

$$\frac{dp}{dx} = 2\left(1 - \frac{36}{x^2}\right)$$

$$\text{put } \frac{dp}{dx} = 0 \text{ gives}$$

$$x = 6 \text{ (neglecting negative value)}$$

$$\text{now } \frac{d^2P}{dx^2} = 2\left[-\frac{(-2)36}{x^3}\right]$$

$$\text{at } x = 6, \left. \frac{d^2p}{dx^2} \right|_{x=6} = \frac{2}{3} > 0$$

Which shows that perimeter is minimum,

$$\text{For } x = 6, \text{ we get } y = \frac{36}{6} \Rightarrow y = 6$$

So the rectangle having area 36cm^2 will have minimum perimeter if length and breadth are 6cm and 6cm .

Q.7 A box with a square base and open top is to have a volume of 4 cubic dm. Find the dimensions of the box which will require the least material.

Solution:

Let h be the height of the box and x be the side of its square base, then

$$v(\text{volume}) = x^2h$$

$$\Rightarrow x^2h = 4$$

$$\text{or } h = \frac{4}{x^2}$$

if s is the surface area, then

$$s = x^2 + 4xh$$

$$s = x^2 + \frac{16}{x} \quad \left(\because h = \frac{4}{x^2} \right)$$

Differentiate w.r.t “ x ”,

$$\frac{ds}{dx} = 2x - \frac{16}{x^2}$$

$$\text{Put } \frac{ds}{dx} = 0 \text{ gives,}$$

$$2x - \frac{16}{x^2} = 0$$

$$2x^3 - 16 = 0$$

$$x^3 - 8 = 0$$

$$\Rightarrow x = 2 \text{ (real value only)}$$

$$\text{Now } \frac{d^2s}{dx^2} = 2 + \frac{2(16)}{x^3}$$

$$\text{at } x = 2$$

$$\left. \frac{d^2s}{dx^2} \right|_{x=2} = 2 + \frac{2(16)}{2^3} > 0$$

So the required surface area is

minimum at $x = 2$.

So the required dimensions are 2dm, 2dm and height 1dm.

Q.8 Find the dimensions of a rectangular garden having perimeter 80, if its area is to be maximum.

Solution:

Let x, y be the length and breadth of that rectangular garden,

$$\text{Then its perimeter } P = 2x + 2y = 80$$

$$\Rightarrow x + y = 40$$

$$y = 40 - x$$

Now if A be the area of rectangular garden then $A = xy$

$$\text{or } A = x(40 - x)$$

Differentiate w.r.t “ x ”,

$$\frac{dA}{dx} = 40 - 2x \text{ and } \frac{d^2A}{dx^2} = -2$$

$$\text{Put } \frac{dA}{dx} = 0$$

$$40 - 2x = 0$$

$$\Rightarrow x = 20$$

$$\text{at } x = 20 \quad \frac{d^2A}{dx^2} = -2 < 0$$

so area will be maximum

So the dimensions are 20cm and 20cm.

Q.9 An open tank of square base of side x and vertical side is to be constructed to contain a given quantity of water. Find the depth in terms of x if the expense of lining the inside of the tank with lead will be least.

Solution:

Let the given quantity of water be ‘ q ’ cubic unit and ‘ h ’ be the depth of the tank, having square base of length x .

$$\text{So the volume of the tank} = x^2h$$

$$\text{So } q = x^2h \Rightarrow h = \frac{q}{x^2}$$

If S be the surface area of inside of the tank, then $s = x^2 + 4xh$

$$\text{or } s = x^2 + 4x \left(\frac{q}{x^2} \right)$$

$$s = x^2 + \frac{4q}{x}$$

Differentiate w.r.t “ x ”

$$\frac{ds}{dx} = 2x - \frac{4q}{x^2} \text{ and } \frac{d^2s}{dx^2} = 2 + \frac{8q}{x^3}$$

$$\text{Put } \frac{ds}{dx} = 0 \text{ gives } 2x - \frac{q}{x^2} = 0$$

$$\Rightarrow q = \frac{x^3}{2}$$

$$x^3 = 2q \Rightarrow x = \sqrt[3]{2q}$$

$$\text{at } x = \sqrt[3]{2q}$$

$$\frac{d^2s}{dx^2} = 2 + \frac{8q}{x^3}$$

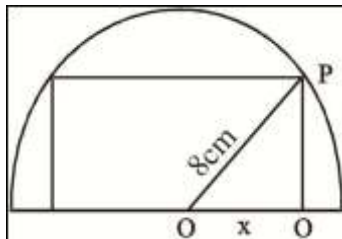
$$\frac{d^2s}{dx^2} = 2 + \frac{8q}{2q} > 0 \quad (\because x^3 = 2q)$$

$$\frac{d^2s}{dx^2} > 0$$

so s is minimum (as required)

$$\text{thus for least expense } h = \frac{q}{x^2} = \frac{x}{2}$$

Q.10 Find the dimensions of the rectangle of maximum area which fit inside the semi-circle of radius 8cm as shown in the figure.



Solution:

Let P be the point on rectangle as shown in figure. Taking O as centre of semi-circle on the origin, the point P will be $(x, \sqrt{64 - x^2})$, if the length of rectangle is taken to be $2x$.

Let A be the area of rectangle, then

$$A = 2x\sqrt{64 - x^2}$$

Differentiate w.r.t ' x '.

$$\frac{dA}{dx} = 2x \frac{1(-2x)}{\cancel{2}\sqrt{64 - x^2}} + \sqrt{64 - x^2} \quad (2)$$

$$= 2\sqrt{64 - x^2} - \frac{2x^2}{\sqrt{64 - x^2}}$$

$$= \frac{2(64 - x^2) - 2x^2}{\sqrt{64 - x^2}}$$

$$\text{Put } \frac{dA}{dx} = 0 \Rightarrow \frac{128 - 2x^2 - 2x^2}{\sqrt{64 - x^2}} = 0$$

$$128 - 4x^2 = 0$$

$$x^2 = 32$$

$$x = 4\sqrt{2}$$

(neglecting negative value of x)

$$\text{Now } \frac{d^2A}{dx^2} = \frac{4x(x^2 - 96)}{(64 - x^2)^{\frac{3}{2}}}$$

(after simplification)

$$\text{At } x = 4\sqrt{2}$$

$$\left. \frac{d^2A}{dx^2} \right|_{x=4\sqrt{2}} = \frac{16\sqrt{2}(-64)}{(32)^{\frac{3}{2}}} < 0$$

So the area of rectangle will be maximum if $x = 4\sqrt{2}$.

Q.11 Find the point on the curve $y = x^2 - 1$ that is closest to the point $(3, -1)$

Solution:

Let ℓ be the distance between the point $(3, -1)$ and a point (x, y) on the curve $y = x^2 - 1$

$$\text{Then } \ell = \sqrt{(x-3)^2 + (y+1)^2}$$

$$= \sqrt{(x-3)^2 + (x^2 - 1 + 1)^2}$$

$$\ell = \sqrt{(x-3)^2 + x^4}$$

The minimum value of ℓ depends on the function

$$\ell = (x-3)^2 + x^4$$

Differentiate w.r.t “ x ”

$$\frac{dl}{dx} = 2(x-3) + 4x^3 \text{ and } \frac{d^2l}{dx^2} = 12x^2 + 2$$

$$\text{Put } \frac{dl}{dx} = 0,$$

$$4x^3 + 2x - 6 = 0$$

$$(2x^2 + 2x - 3)(x-1) = 0$$

$$2x^2 + 2x - 3 = 0 \text{ gives no real roots}$$

$$\text{So } x-1=0 \Rightarrow x=1$$

$$\text{At } x=1 \quad \frac{d^2l}{dx^2} = 12(1)^2 + 2 = 14$$

$$\frac{d^2l}{dx^2} > 0$$

So l has minimum value at $x=1$

$$\text{Put } x=1 \text{ in } y = x^2 - 1$$

$$y = 0$$

Hence the required point on the curve is $(1,0)$

Q.12 Find the point on the curve

$y = x^2 + 1$ that is closest to the point $(18,1)$

Solution:

let l be the distance between the point $(18,1)$ and a point (x,y)

on the curve $y = x^2 + 1$ then.

$$l = \sqrt{(x-18)^2 + (y-1)^2}$$

$$= \sqrt{(x-18)^2 + (y-1)^2}$$

$$(\because x^2 + 1 = y)$$

$$= \sqrt{(x-18)^2 + x^4}$$

The minimum value of l depends on the function

$$l = (x-18)^2 + x^4$$

Differentiate w.r.t “ x ”

$$\frac{dl}{dx} = 2(x-18) + 4x^3 \text{ and } \frac{d^2l}{dx^2} = 12x^2 + 2$$

$$\text{Put } \frac{dl}{dx} = 0$$

$$2x^3 + x - 18 = 0$$

$$(x-2)(2x^2 + 4x + 9) = 0$$

$$2x^2 + 4x + 9 = 0 \text{ gives complex roots}$$

$$\text{so, we must have } x-2=0 \Rightarrow x=2$$

$$\text{At } x=2 \quad \frac{d^2l}{dx^2} = 12(2)^2 + 2 = 50$$

So l has minimum value at $x=2$

put $x=2$ in $y = x^2 + 1$ we get

$$y = 2^2 + 1 = 5$$

Hence the required point on the curve is $(2,5)$.